

SHORT COURSE

Data Analytics for Business and Healthcare

A hands-on introduction to exploring real-world data with Python & Google Colab, no coding background required

By Dr. Norliana & Dr. Chen Xinyuan



Welcome!

1

Foundations

What data analytics is, and why it matters in business & healthcare

2

Tools

Meet Python, Google Colab, and the analytics workflow

3

Hands-On Lab

Explore a real COVID-19 Malaysia dataset step-by-step

4

Kahoot Activities

Two rounds of fun, interactive quizzes to test what you learn

Learning Objectives

By the end of this short course, you will be able to:



Explain what data analytics is and why it drives decisions in business and healthcare



Describe the 5-step data analytics process: collect, clean, explore, analyze, communicate



Run simple Python code in Google Colab with zero prior coding experience



Load, clean, and explore a real dataset using pandas



Create charts to visualize trends and communicate insights



Apply what you learn about COVID-19 data to real business and healthcare scenarios

How Today Is Structured

1 hr 30 min

Lecture + Hands-On Lab

Concepts explained simply, then applied live in Google Colab using the COVID-19 Malaysia dataset

15 min

Kahoot Activities

Two interactive quiz rounds on your phone — no app download, just a game PIN

Total course time: 1 hour 45 minutes

What Is Data Analytics?

“

Data analytics is the process of examining raw data to find patterns, draw conclusions, and support better decisions.

In short: turning numbers into knowledge.

Data

Raw facts and figures : numbers, dates, text, images

Analytics

Techniques used to examine and interpret that data

Insight

The useful meaning we extract to guide action

FOUNDATIONS

What Is Data?



Structured Data

Neatly organized in rows and columns : spreadsheets, databases, billing records.



Unstructured Data

Text messages, photos, videos, social posts, no fixed format at all.



Semi structured

Combination of structured + unstructured.



Data Analytics in Business

Retail

Analyzing shopping baskets to design promotions and store layout

Banking

Detecting unusual transactions to prevent fraud

Marketing

Measuring which campaigns bring the best return on investment

Logistics

Optimizing delivery routes to reduce fuel cost and delays

Human Resources

Identifying factors linked to staff turnover

Data Analytics in Healthcare

Epidemiology

Tracking how a disease spreads through a population, like COVID-19

Hospital Management

Predicting bed and ICU demand to plan staffing

Public Health Policy

Evaluating the impact of vaccination or lockdown measures

Clinical Care

Spotting patients at risk based on health records

Pharma

Monitoring drug safety and effectiveness after launch

The Data Analytics Process

1

Collect

Gather data from reliable sources

2

Clean

Fix errors & missing values

3

Explore

Look at patterns & summary stats

4

Analyze

Apply methods to find insight

5

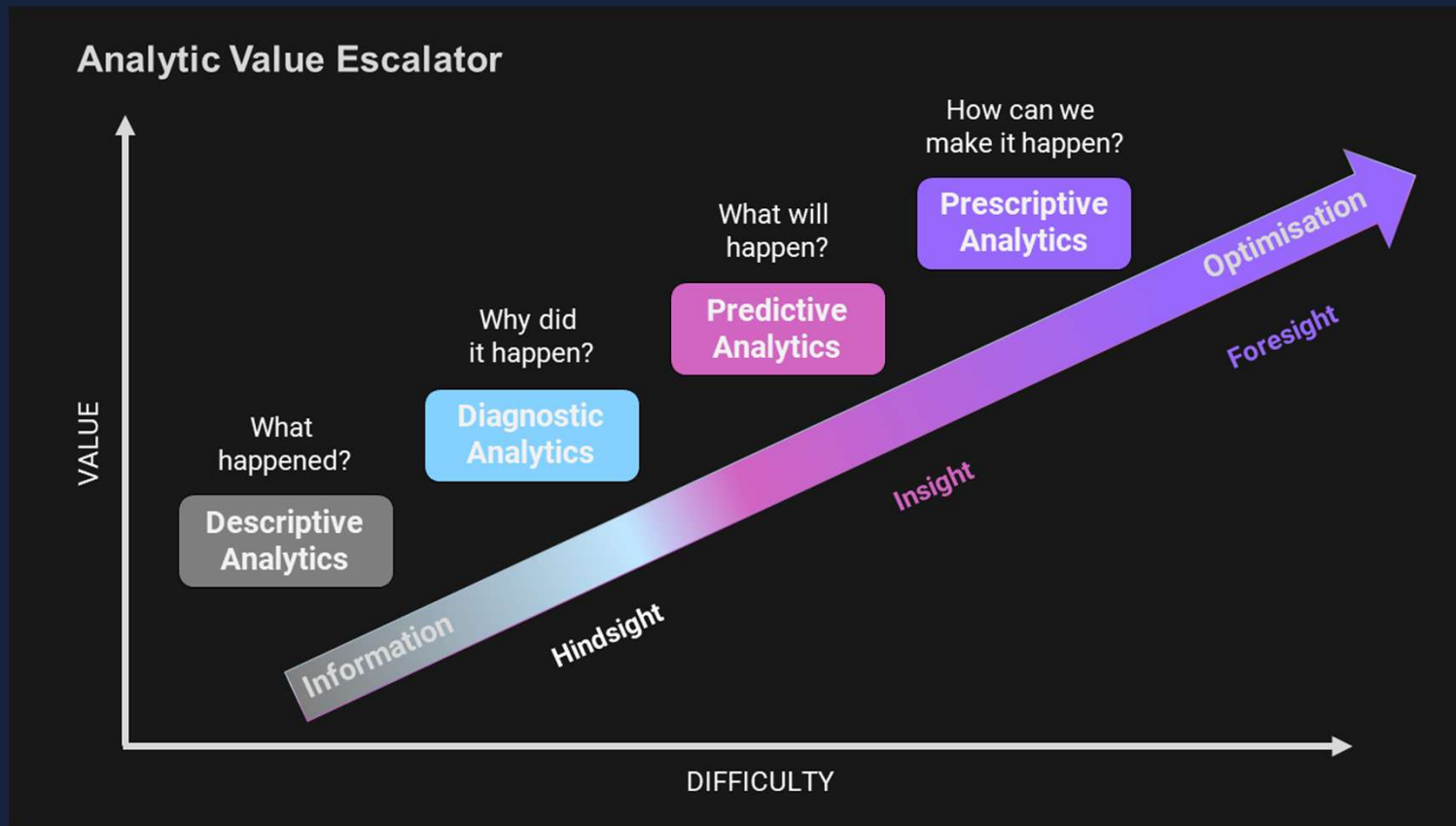
Communicate

Share findings clearly with charts

We will walk through all 5 steps live in today's hands-on lab.

The Four Types of Analytics

Each type builds on the last from hindsight to foresight to action



The Analytics Toolkit

Tools everyone should recognize

Tool	Best For	Key Feature
Excel	Quick reports & simple models	Familiar spreadsheets & PivotTables
SQL	Querying company databases	The universal language of data
Python / R	Advanced statistics & ML	Full control over custom analysis
Power BI / Tableau	Interactive dashboards	Drag-and-drop data visualization

Data Visualization I

Five charts, two data types, the foundation for everything

Pie Chart

% composition, few categories



Pie Chart

Bar Graph

Compare across categories



Bar Graph

Histogram

Distribution of grouped numeric data



Histogram

Freq. Polygon

Trend shape across class intervals



Freq. Polygon

Stem-and-Leaf

Every raw value, distribution shape



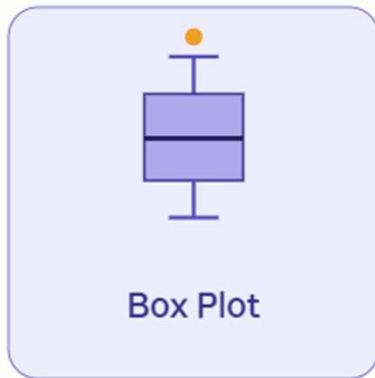
Stem & Leaf

Data Visualization II

Four chart types that unlock deeper interpretation

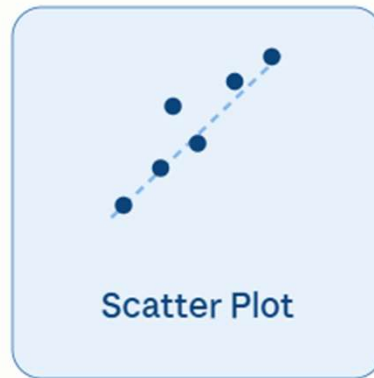
Box Plot

Spread, median & outliers



Scatter Plot

Relationship between 2 variables



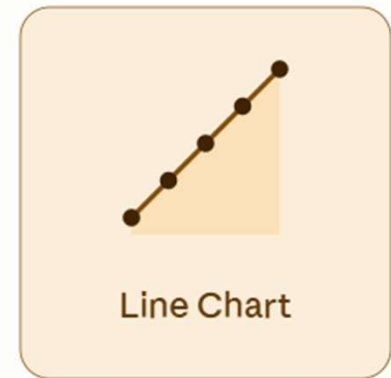
Heatmap

Correlation across many variables



Line Chart

Trends & change over time



Data Types: The Foundation of Every Chart

Before picking a chart, you must know what kind of variable you're looking at

CATEGORICAL (QUALITATIVE)

Describes a quality or category : cannot be measured numerically.

Nominal

Categories with NO natural order

e.g. Department, Gender, Blood Type

Ordinal

Categories WITH a meaningful order

e.g. Grade (A/B/C), Satisfaction Level

NUMERICAL (QUANTITATIVE)

Measured on a numeric scale : can be counted or calculated.

Discrete

Countable whole numbers only

e.g. App Downloads, Number of Siblings

Continuous

Any value within a range

e.g. Study Hours, Height, Score



KAHOOT ACTIVITY #1

Data Analytics Basics : Quick Quiz

Grab your phone • Go to kahoot.it • Enter the Game PIN on screen

10 MINUTES • 10 QUESTIONS

Meet Python

The programming language we will use : beginner-friendly & world's most popular for data analytics

Easy to read

Python code looks close to plain English

Free & open-source

No cost, no license required

Huge community

Millions of users and ready-made tools

Built for data

Powerful libraries like pandas and matplotlib

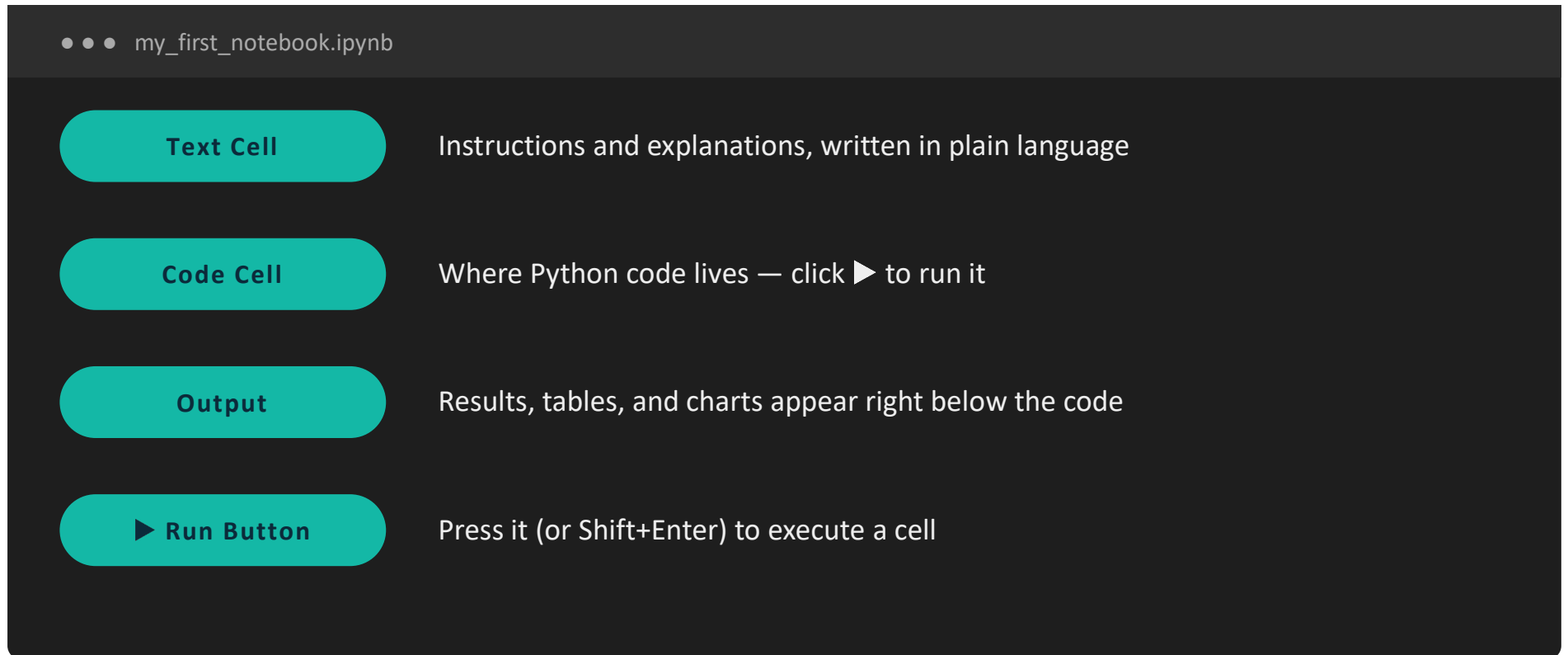
What Is Google Colab?

A free website where you can write and run Python code in your browser

colab.research.google.com

- No installation needed, just a Google account
- Runs on Google's computers, not your own laptop
- Great for learning and sharing analysis
- This is exactly what we'll use today's lab notebook in

A Quick Tour of Colab



• • • my_first_notebook.ipynb

Text Cell	Instructions and explanations, written in plain language
Code Cell	Where Python code lives — click ► to run it
Output	Results, tables, and charts appear right below the code
► Run Button	Press it (or Shift+Enter) to execute a cell

Our Two Main Tools Today



pandas

Library for working with tables of data

Used to load our COVID-19 dataset, view rows and columns, clean missing values, and calculate summary statistics like totals and averages.



matplotlib

Library for creating charts

Used to turn columns of numbers into line charts and bar charts, so trends in the data become easy to see at a glance.

CASE STUDY

COVID-19 in Malaysia

Real, open data from Malaysia's Ministry of Health (MoH), the same data used by health authorities and analysts during the pandemic

- ✓ It's relevant, a shared national experience for all Malaysians
- ✓ It's real, official public data, not made-up numbers
- ✓ It shows both angles, public health AND economic impact
- ✓ It's a perfect size for beginners to explore in one hour

About the Dataset

File: covid19_malaysia_cases.csv | 730 daily records: 1 Jan 2021 – 31 Dec 2022

date

The calendar date

cases_new

Newly confirmed COVID-19 cases that day

cases_recovered

Patients who recovered that day

cases_active

Total patients currently infected

deaths_new

COVID-19 deaths recorded that day

vax_cumul_fully_vaccinated

Running total of fully vaccinated people

Data Source & Ethics

Official Source

Ministry of Health Malaysia (MoH) published openly on GitHub: github.com/MoH-Malaysia/covid19-public.
Updated daily throughout the pandemic for public use.

A Note on Data Ethics

- This dataset contains only aggregated national counts, no individual is identifiable
- Health data at the individual level must always be handled with strict privacy safeguards
- Good analysts always check: is this data public, consented, and used responsibly?

Upload & Load the Dataset

```
import pandas as pd

from google.colab import files

uploaded = files.upload()    # choose covid19_malaysia_cases.csv
df = pd.read_csv('covid19_malaysia_cases.csv')
```

We import pandas (our data toolkit), then upload the dataset file straight from our computer using Colab's file picker. `read_csv()` loads it into a table called a DataFrame — think of it like an Excel sheet inside Python, stored in the variable `df`.

Note: the notebook breaks this into a few small, separately-run cells for clarity — this slide shows the combined logic.

First Look at the Data

```
df.head()  
  
df.shape  
  
df.info()
```

`head()` shows the first 5 rows so we get a quick preview. `shape` tells us the number of rows and columns. `info()` lists every column and its data type — this is always the very first thing an analyst does with new data.

Data Diagnosis & Cleaning Basics

```
df.isnull().sum()  
  
df['date'] = pd.to_datetime(df['date'])
```

`isnull().sum()` counts missing values in every column — real-world data is rarely perfectly complete. We also convert the date column to a proper date type so Python understands it as a timeline, not just text.

Descriptive Statistics

```
df.describe()  
  
df['cases_new'].mean()  
  
df['cases_new'].max()
```

`describe()` instantly gives us the average, minimum, maximum, and spread for every numeric column. This answers simple but powerful questions: what was a typical day like, and what was the worst day?

Visualizing the Trend of Daily Cases

```
import matplotlib.pyplot as plt

plt.plot(df['date'], df['cases_new'])

plt.title('Daily New COVID-19 Cases in Malaysia')

plt.show()
```

A single line chart reveals what hundreds of rows of numbers cannot: the shape of the pandemic — the waves rising and falling over 2021 and 2022 — instantly visible to the human eye.

Comparing Cases vs. Deaths

```
plt.plot(df['date'], df['cases_new'], label='New Cases')  
plt.plot(df['date'], df['deaths_new']*100, label='Deaths (x100)')  
plt.legend()  
plt.show()
```

Plotting two lines together lets us compare patterns — did deaths rise and fall along with cases, or with a delay? This is exactly how public health teams monitored the severity of each wave.

Vaccination Progress Over Time

```
plt.plot(df['date'], df['vax_cumul_fully_vaccinated'])  
plt.title('Cumulative Fully Vaccinated Population')  
plt.show()
```

A cumulative (running total) chart always climbs upward. Its shape shows us the speed of the vaccine rollout — a steep climb means rapid uptake, a flattening curve means the rollout is slowing down.

Finding the Peak: Identifying the Waves

```
peak_day = df.loc[df['cases_new'].idxmax()]\nprint(peak_day)
```

`idxmax()` finds the row with the highest value — instantly pinpointing Malaysia's worst day for new cases (around early March 2022, over 33,000 cases in a single day). Analysts always ask: what happened on that day, and why?

From COVID Data to Business Data

The exact same skills apply directly to everyday business questions:

cases_new over time



Daily sales over time

cases_active



Number of open customer support tickets

Peak day analysis



Identifying your best-selling day or campaign spike

Cumulative vaccination



Cumulative product sign-ups or app downloads

From COVID Data to Hospital Planning

The same trend-and-peak analysis directly supports resource planning:



Forecast Demand

Rising case trends give hospitals early warning to prepare beds and staff



Allocate Resources

Peak analysis shows exactly when ICU capacity was most strained



Evaluate Interventions

Comparing trends before/after vaccination shows what worked

Common Pitfalls to Avoid



Correlation $\neq$ causation

Two trends moving together doesn't mean one causes the other



Garbage in, garbage out

Bad or incomplete data leads to misleading conclusions



Cherry-picking

Don't only show the data points that support your idea



Ignoring context

Always ask what else was happening (policy, season, events)



Privacy carelessness

Never expose personally identifiable health or customer data



KAHOOT ACTIVITY #2

Interpret the Data : Scenario Challenge

Read each mini case, then pick the best data-driven decision

20 MINUTES • 8 SCENARIOS

Key Takeaways

- 1 Data analytics turns raw numbers into decisions in any industry
- 2 The process is always: Collect → Clean → Explore → Analyze → Communicate
- 3 Python + Google Colab let anyone start analyzing data for free, in the browser
- 4 pandas explores tables of data; matplotlib turns them into charts
- 5 The COVID-19 Malaysia case study skills apply directly to business & healthcare

Your Data Analytics Cheat Sheet

```
pd.read_csv(url)
```

Load a dataset

```
df.head()
```

Preview first rows

```
df.info()
```

Check columns & types

```
df.describe()
```

Get summary statistics

```
df['col'].mean()
```

Average of a column

```
df['col'].max()
```

Highest value

Where to Go From Here

1

Practice

Re-run the Colab notebook and try changing the chart to a different column

2

Explore more data

data.gov.my : Malaysia's Open Data portal has many free datasets

3

Learn more Python

Free courses: Kaggle Learn, freeCodeCamp, Google's own Python class

4

Apply it at work

Pick one report at your workplace and try to recreate it using pandas

Terima Kasih!

Thank you : Questions & Discussion

CONTACT:

- 1) DR. NORLIANA MUSLIM (norliana@unikl.edu.my)
- 2) DR. CHEN XINYUAN (chen.xinyuan@unikl.edu.my)



We personally conduct a full, comprehensive training program in Data Analytics for Business and Healthcare, which goes significantly beyond this introductory session. Participants interested in developing complete, job-ready data analytics skills are warmly invited to join the full program.

Everything You're Taking Home



Slide Deck (PPTX)

This presentation, for your own review



Case Study Dataset (CSV)

COVID-19 Malaysia data, ready to reuse



Hands-On Lab Manual

Step-by-step written guide for the lab



Kahoot Quiz Activities

Both question banks, ready to load into Kahoot



Python Notebook (Colab)

Fully working, ready to run in Google Colab